

Case based antibiotic selection for various bovine diseases: Where evidence-based veterinary medicine meets spectrum of care

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Abstract

In clinical practice, veterinarians are routinely called on to make antimicrobial decisions. Given the wide range of bovine diseases, scope of available antimicrobials, and sometimes lack of evidence-based veterinary medicine reviews that align with a case, veterinarians can feel challenged deciding which drug to use. This paper reviews various factors that should be considered and communication tools that could be used to formulate an acceptable treatment plan with clients.

Key words: evidence-based medicine, spectrum of care, antimicrobial stewardship

Introduction

How do veterinarians choose which medication to treat a particular disease? There are well written textbooks and peer-reviewed journal articles that cover bovine diseases including pathophysiology, causative agents, preventive measures and multiple possible treatments.^{5,9,13} However, in addition to being familiar with the basic information about these common diseases, veterinarians of all years of experience are now called on to balance antimicrobial stewardship (AS) with evidence-based veterinary medicine (EBVM) and to bring these treatment solutions to clients at the speed of commerce. The intent of this paper is to review what should be considered before any selection of antimicrobial therapy. After that brief review, the discussion will focus on some of the strengths and limitations of EBVM. Finally, the discussion will touch on how to use spectrum of care (SoC) and communication tools to provide therapy for bovine patients while honoring antimicrobial stewardship.

Disease pathophysiology, diagnosis and prevention

Before choosing a drug, the first step is to establish a working diagnosis. Veterinarians should perform a complete physical exam to identify the problem(s) and possible disease(s) that need to be treated and establish a veterinary client patient relationship.⁶ Well written textbooks and peer-reviewed journal papers should give a clear definition of disease with objective physical exam findings to help you rule in/out a disease. These resources should also be used to review underlying pathophysiology and risk factors to obtain the long-term farm goal of preventing disease in other animals.⁸ Equally important, veterinarians should use communication tools to gather history and relevant production information during the exam to understand the clients' goals.¹² In some cases, the patient will present with symptoms that are classic for a disease (e.g. uterine prolapse). Still, clinicians should make every effort to come up with a reasonable list of differentials and necessary diagnostics

before presuming there is only one disease and drawing anything into a syringe.

Antimicrobial selection

The same resources that help define the diseases may also include treatment options for that disease. The *Compendium of Veterinary Medicine* also has the labels for all approved drugs that will show what diseases the drug has been approved for as well as the proper dosage and withdrawal dates.¹⁷ Despite this, there are common diseases such as navel infections in a newborn calf that have no labeled treatments. Michael Apley DVM, MS PhD, ACVIM at Kansas State University has put together a selection of resources on antimicrobials and AS that can be used to help veterinarians and veterinary students alike to consider what properties of a drug make it potentially "right drug for the right bug" when there is no labeled product.^{1,2}

There are also legal aspects that should be considered before treatment is given. Practitioners must be aware of state and federal laws and regulations concerning administering, dispensing and prescribing drugs. These laws and regulations include the Animal Medicinal Drug Use Clarification Act (AMDUCA) for extra label use of drugs in cattle. There is also FDA's Guidance for Industry #213 concerning the Veterinary Feed Directive that forbids extra label use of feed additives.

Finally, some concepts that were once emphasized for the drug selection process have fallen out of favor. For instance, traditionally, veterinarians were encouraged to choose bactericidal over bacteriostatic for treatment but in 2018 Wall-Dicker, et al. took a look at this dogma in a systematic human trial review. In summary "virtually all available data from high-quality randomized clinical trials demonstrate there is no intrinsic superiority of bactericidal compared to bacteriostatic agents. Other drug characteristics such as optimal dosing pharmacokinetics and tissue penetration may be more important efficacy drivers."¹⁶

Antimicrobial stewardship

An even more recent and heavily discussed topic in veterinary medicine is antimicrobial stewardship (AS). In the American Association of Bovine Practitioners 2017 guidelines for implementing antimicrobial stewardship, AS is defined as the commitment to reducing the need for antimicrobial drugs by preventing infectious disease in cattle, and when antimicrobial drugs are needed, a commitment that antimicrobials are used appropriately to optimize health and minimize selection for antimicrobial resistance. This short positional paper has key areas to cover to make sure that we do not contribute to antimicrobial resistance however it does not dictate which drugs to use except to say that veterinarians should continuously seek new information about the use of antimicrobial drugs.

It is also noteworthy under AS to mention the shift that has occurred recently on the interpretation of resistance studies coming out of the university diagnostic labs when selecting antimicrobials to use. Magstadt et al. sounded the alarm that “previous antimicrobial treatment may have a profound effect on antimicrobial susceptibility testing. Summarized susceptibility results from diagnostic laboratories should not be used to make generalized statements regarding trends in antimicrobial resistance without providing context regarding antimicrobial treatment history.”¹¹

Evidence-based veterinary medicine’s role in antimicrobial selection

When called under AS to justify the need and choice of antibiotics, veterinarians should be driven to find evidence to support their selection. Evidence-based medicine has been described as “the integration of best research evidence with our clinical expertise and our patients’ unique values and circumstances”. Dr Vanderweerd and Buczinski go further to define evidence-based veterinary medicine as “the use of accurate, informative and clear summarized information of high-level research studies that is obtained quickly via free access databases available via the internet and that is provided by a proactive veterinary scientific community in search of transparency, accountability and evidence.”¹⁵ They see EBVM, especially in bovine practice, as an information tool that does not replace experience but helps to improve background knowledge and solve clinical foreground questions.

The pursuit of more evidence is noble. However, there are numerous papers written about the limitations of EBVM in veterinary medicine, not the least of which is limited time to consult databases.³ This is even more difficult because not all databases are freely available and search engines may not be adequate for all veterinary disciplines. Some topics contain exhaustive results while others are extremely limited. Another relatable limitation to the practice of EBVM is the need for veterinarians to develop new skills in seeking and appraising evidence in literature. Some continuing education, especially in the form of literature reviews in podcasts, can help veterinarians work on these skills. Finally, there are often limitations in observational cohort studies in bovine medicine. Common limitations in study characteristics might include selecting the herds or the cows within a herd to enroll based on convenience, lack of the ability to double blind treatments and observations and the inherent variation in subjective data if that is part of a case definition (e.g. uterine discharge color and odor for a case definition of metritis.) Another struggle with EBVM medicine is that it can be applied most accurately when looking at populations in the study that are similar to our populations in private practice. There is a reason it is called mixed animal practice. Not only do veterinarians deal with a variety of species throughout the year they also deal with a variety of breeds, ages, weights and management styles which may not be reflected in the well-written study.

Still, it is reasonable to take small bites of EBVM as it comes out and think about if this could apply to the population that is in front of the veterinarian.

Spectrum of care

Finally, before a veterinarian can choose the right drug based on previously discussed points of consideration, the veterinarian should understand the concept of spectrum of care (SoC).

At its core, SoC is recognition that acceptable veterinary care isn’t a fixed point, but rather a continuum tailored to meet the specific medical needs of each animal while simultaneously considering a holistic picture of the associated client circumstances.⁴ A paper written by clinicians at The Ohio State remarks how SoC can provide a continuum of acceptable care that considers available EBVM while remaining responsive to clients’ expectations and financial limitations, thereby successfully serving an economically diverse clientele.⁷ Ideally veterinarians should be able to provide several evidence-based options for care that encompass a wide variety of diagnostic and treatment approaches for the client and patient. Surveys show that when students transition from clinical year to general practice that they may not feel comfortable offering Plan B, plan C or plan D approaches to care. Rather they may have been instructed that a treatment plan is “all-or-nothing”, and if the client doesn’t accept it all, there is nothing they can do for the patient. In a 2018 survey by Kipperman, et al. 52% of respondents acknowledged that they experienced an ethical dilemma as a DVM at least once a week related to this point.⁷ While this paper was in a small animal clinical setting it is reasonable to assume that bovine veterinarians face these challenges as well as imposter syndrome when clients look to the doctor to “do the right thing” for their animal.

Putting it into practice

How do veterinarians choose which medication to treat a particular disease? A better question may be “how does a veterinarian ever come up with a recommendation given all these considerations?” It can seem like a daunting task. The discussions about EBVM, AS and SoC, simply show that veterinary medicine has room to continue to grow and evolve. If a veterinarian feels like they don’t have all the answers, a quick literature review on any given disease will reveal very little top tier evidence to point to anyone having it 100% figured out. So how does a busy veterinarian get anything done? Compared to our companion animal colleagues, bovine veterinarians have fewer clinical textbooks and workshops covering client communication skills. However, many of the tools used in other areas of veterinary medicine, such as the value matrix, could also be utilized in bovine practice to solve this dilemma. The value matrix is a universal way to reflect back values that you have gathered while taking history and be able to deliver concise treatment options that will help veterinarians build rapport with their clients. This value matrix is a visual tool that can be used to present each diagnostic and treatment option, discuss the advantages and disadvantages, approximate the cost, return on investment (and relevant withholds associated with treatments) and make a shared decision with the client. Perhaps the best way to tie all these concepts together is with a case example

Case example: Beef cow with retained placenta and metritis

A beef client makes an appointment to “have their cow cleaned out” that still has her placenta 3 days after delivering live twin heifer calves. Fortunately, by the time you arrive, the placenta has dropped, but the client has concerns about possible infection. This is the last cow from dad’s old herd (they plan to bury her on the farm someday), and they are really hoping that she can make enough milk to feed both twins, so they don’t have to graft her calf to another cow or have a bottle calf. They will be keeping her up close to the working facilities to watch her and

the calves. On physical exam of this 7-year-old black Angus cow, you note a rectal temperature of 103.5 °F (39.7 °C) and brown, fetid vaginal discharge. The manure is normal, the rumen contracts 3 times/minute, and there are no pings on the left or right sides of the abdomen. The heart and lungs auscultate normal; the body condition score is 6/9 and the udder has normal conformation with normal appearing milk in all 4 quarters. She is quiet, alert, responsive, but her eyes are moderately sunken, and her abdomen is flat-sided.

If you have 15 more appointments that day, you might be inclined to roll the window down as you pass by and say “good, she cleaned on her own.” But taking time to do the physical exam shows clinical signs of puerperal metritis. A review of literature suggests that the most common pathogens identified in this disease are *Fusobacterium necrophorum*, *Escherichia coli*, *Trueperella pyogenes*, *Prevotella* spp.⁵ What is the therapy of choice? Literature review searches of metritis tend toward dairy animals instead of beef. They also often discuss the outcomes of treat vs. no treat rather than discuss the “gold standard” drug to give. There are even well-written papers that suggest no significant clinical difference (other than milk production) between treating metritis with an antibiotic vs. just supportive care.⁸ But your diligent history skills picked up on “best cow”, “dying on the ranch” and “don’t want bottle calves” and your physical exam leads you to believe that she is run-down by this ongoing disease process. Telling this client that antibiotic therapy may not have a statistical return on investment will not align to the client’s stated goals.

If we are going to give antimicrobials, what drives our decision when there is no one “gold standard”? Dr. Coe discusses drawing a 2x2 value matrix on a piece of paper that has patient and client benefits listed across the top and treatment options listed down the side.⁴ While filling in the chart in front of the client on a feed sack, the veterinarian can review the pros and cons of various treatments utilizing different symbols. This helps clarify which treatment may be best for this client (Table 1).

At the end of the visit, the client and veterinarian might agree to do a dose of Excede®,¹⁷ supportive care consisting of an oral drench of calcium propionate and vitamin mixture, and leave a bag of oral drench for the owners to mix into her water tank tomorrow. She will need one more dose of Excede in 72 hours in the opposite ear. The client chose these treatment options so the cow does not develop aversion to the headgate that they need to put her in to ensure the calves are nursing. From a welfare standpoint, you could consider flunixin meglumine topically for pain and inflammation as well.

At the end of the day, there are multiple correct antimicrobial options. Supply chain shortages may change what any veterinarian selects on any given day. It is noteworthy that some options are not on the matrix. Antimicrobial stewardship would suggest that picking multiple antibiotics at the same time is not the correct option. Also, while there are a few scientific articles that discuss various uterine infusions that have been tried, you will notice it is not on this chart.⁸ That is because 25 years of dealing with massive adhesions after herdsmen have attempted uterine infusions through a partially closed cervix has

Table 1: Value matrix of treatment options for beef cow with puerperal metritis.

Treatment options	# of treatments	Patient and client benefits			
		Ease of treatments	Meat withdrawal	Cost of treatment	Milk production and repro prognosis
No treatments at all	0	++++	0 days	0	- -
Supportive care (oral calcium/electrolyte drench)	1-3	/ +	0 days	\$	-
Non- antibiotic therapies (Oxytocin ¹ +/- flunixin meglumine transdermal ²)	1	+++	0 / 8 days	\$	-
Ampicillin 25g ³	3-4	+	6 days	\$\$	++
Ceftiofur ⁴	2	++	13 days	\$\$\$\$	++
Ceftiofur RTU ⁵	5	+	4 days	\$\$\$\$	++
Oxytetracycline 200 mg/ml ⁶	2-4	+	28 days	\$	++
Penicillin G 300,000 IU/ml ⁷	5	+	20+ days, consult FARAD ⁸	\$\$	++

¹ 4IU intramuscularly (IM)

² 1.5mg/lb (3.3mg/kg) topically

³ 3 to 5 mg/lb (1.6 to 2.27 mg/kg) q 24 hr IM

⁴ 3mg CE/lb (6.6mgCE/kg) (1.5ml/100lbs) q 72 hours base of ear - Excede®¹⁷

⁵ 1 mg CE/lb (2.2 mg CE/kg) (2ml/100lbs) subcutaneously (SQ)

⁶ 5mg/lb (2.27mg/kg) q 24 hr SQ

⁷ Typical extra label drug usage dose is 15,000IU/lb (33,0000 IU/kg) q24hr IM⁵

⁸ Food Animal Residue Avoidance Databank <https://www.farad.org/>

taught this clinician the true meaning of “do no harm”. There is also no oral chlortetracycline (CTC) in free choice mineral option. That’s because of the use of CTC in a free choice mineral for any indication other than anaplasmosis is forbidden by the Veterinary Feed Directive. In addition to the legal aspect, recent studies on plasma concentrations of CTC after cattle were fed products according to label doses cast doubts on our ability to achieve therapeutic plasma levels through any feed substrate.^{10,14}

The point of this exercise is to figure out a short list of things a veterinarian might add to their value matrix (maybe the veterinarian combines supportive care, hormones and anti-inflammatories all under one line) and then go through some options with the client. The client should be involved in a manner that helps them feel comfortable following the veterinarian’s treatment recommendations. Ultimately, clients are looking for the veterinarian to make a recommendation that they can accept or decline based on the withdrawals, price, labor intensity and likelihood of a favorable prognosis. When the doctor shows the pros and cons of different treatment options, the client can be invested in the treatment which ultimately improves compliance as well as strengthens the relationship between client and veterinarian.

Conclusion

It takes time and effort to balance the latest evidence-based medicine and antimicrobial stewardship with the large volume and wide spectrum of bovine patient cases that are seen in private practice. Utilizing information from various peer reviewed journals, continuing education meetings and veterinary podcasts can help the busy practitioner develop different options which may be equally effective. By utilizing a value matrix and other communication tools to explain options to a client, veterinarians can develop a treatment plan that will work for the patient and strengthen client relationships.

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Conflicts of Interest

The authors declare no conflicts of interest. No AI-assisted technologies were used in the generation of this manuscript.

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